

Reasoning:

Lesson 2 has students engaging in scientific argumentation when working to rank evidence from high quality to low quality. This ranking activity has students exploring concepts such as sample size, confounding variables, control groups, subject diversity, and anecdotal evidence. These conversations will reveal students' current ideas of sound scientific methods. The evidence that students explore in this activity relates to vaccination, which explicitly shows students how practices utilized in the scientific classroom transfer to decisions they make in their lives. During partner and class discussions, students will have the opportunity to offer constructive criticism to their peers. Students use information gathered during the class discussion of evidence to develop a list of criteria for good scientific methods. This list of criteria will be edited as additional evidence of good methods comes to light throughout the unit, highlighting the idea that scientific understanding is dynamic and can be adjusted as new evidence is revealed. Time for metacognitive reflection is incorporated at the end of the lesson to increase the likelihood of retention and transfer.

Slides	Notes
DO NOW! LESSON 2 While we spent a bit of time making models yesterday, it takes many years for scientists to develop and agree on models. Why might it take so long? Instructions: Write your thoughts about this in our "Reflections Sheet" (in the tab with your name on it).	Use think-pair-share protocol 💡 Ideas to elevate: • Collection of ample evidence takes time • Feedback must be collected and considered throughout the process
CHARACTERISTICS OF GOOD SCIENTIFIC METHODS Think about the last part of the homework assignment, where you ranked the six pieces of evidence from best to worst. Your rankings and reasons for your rankings give you ideas about what are some good scientific methods for conducting research. Class Task: Now we'll make a list of good scientific methods. 22	Have students compare their rankings with a partner and have a full class discussion. Elicit student ideas on which method criteria are most important when considering evidence (i.e. Is a large sample size more important than a control group?). You may not arrive at an answer, but students may start to see that these considerations are sometimes context specific. 🔍 Once the evidence is discussed, lead a class discussion on criteria for good scientific methods. This list should be visible in the classroom either on poster paper or a digital slide. 🔍 Possible things to include on list: 1. Include comparisons 2. Conduct carefully controlled comparisons. 3. Use larger samples. 4. Use more objective measures (not just subjective feelings but also measuring fevers, etc.) 5. Use proper study duration (the study shouldn't measure improvement at a time after everyone will already recover anyway)

6. Use multiple time measures (record results daily over time)

PLEASE ENCOURAGE STUDENTS TO DO THESE TWO THINGS:

1. Explain their reasons for why particular criteria are important.
2. Explain why particular criteria are or are not found in different pieces of evidence that they looked at.

REVISITING THE CLASS CRITERIA FOR 'GOOD SCIENTIFIC MODELS'

REVISIT CLASS CRITERIA OF GOOD MODELS

1. What were your criteria of good models?
2. Which do you consider the most important criterion and why?
3. Are there improvements we can make to these criteria? Additions?

24

Have students consider part 1 of the homework assignment. Ask students if all pieces of evidence of the same quality. 🔍

Low quality pieces of evidence you can highlight:

- The mere two cases of children who both had the disease and had mosquito bites. This could be just chance; it's a tiny sample.
- Suppose one person tells their doctor that they are convinced they got the disease from a mosquito bite. Do scientists need to consider that evidence?

Ask students: What kind of evidence do we need to consider, then, when we make our models?

- Hopefully, some students will suggest that models need to fit all the good evidence, or all the evidence that isn't bad, or something like that.

Ask them what they think about revising the criterion to accommodate this idea that they are developing.

The revised criteria might be fitting all the good evidence, or the evidence with good methods, or something of that sort. They may suggest different wording. It is important to connect to "good methods" in some fashion, because the next activity (next slide) develops a list of good methods. 🔍🧪

If they agree to a change, revise the class criteria accordingly.